



NCCER Module 22201



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Introduction to Earthmoving





Section Two

Site Excavation Practices

Objective

2. Describe the equipment and methods used in excavating.
 - a. Describe how to measure production for various earthmoving activities.
 - b. Describe the equipment and methods used in loading excavated soil.
 - c. Describe the equipment and methods used in hauling excavated soil.
 - d. Describe the equipment and methods used in dumping excavated soil.
 - e. Describe fill, backfill, and embankment construction.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

2. Identify and select the proper equipment for a given earthmoving operation.



2.0.0 Site Excavation Practices (1 of 4)

Before starting work:

- Contractors need to analyze the job and determine the type of equipment needed.
- Equipment operators should study the plans and stakes carefully.
- A planner, engineer, or supervisor must determine if there are any utilities or other structures buried on the site.



2.0.0 Site Excavation Practices (2 of 4)

Equipment is chosen to match to the type of material to be excavated.

Left: Scraper Right: Excavator



2.0.0 Site Excavation Practices (3 of 4)

Equipment is chosen to match to the type of material to be excavated.

Left: Motor Grader Right: Dozer



2.0.0 Site Excavation Practices (4 of 4)

Equipment is chosen to match to the type of material to be excavated.

Left: Wheel and track loader Right: Backhoe loader



2.1.0 Production Measurement

- Although equipment operators usually do not get involved in the details of planning the overall project work, they need to understand how it is done.
- Equipment operators may provide suggestions or input based on their experience with different types of soils and the handling of different machines under various conditions.



2.1.1 Mass Haul Diagram

- The mass haul diagram is a graphic method of analyzing the movement of soil from one loading area to the dumping site.
- It relates the proposed grade to the existing grade.
- It is useful in determining the location and volume of excavations and the haul distances to points in the fill.
- It shows where to dump surplus excavated material in a **spoils** area rather than haul it to a fill area that is further away.



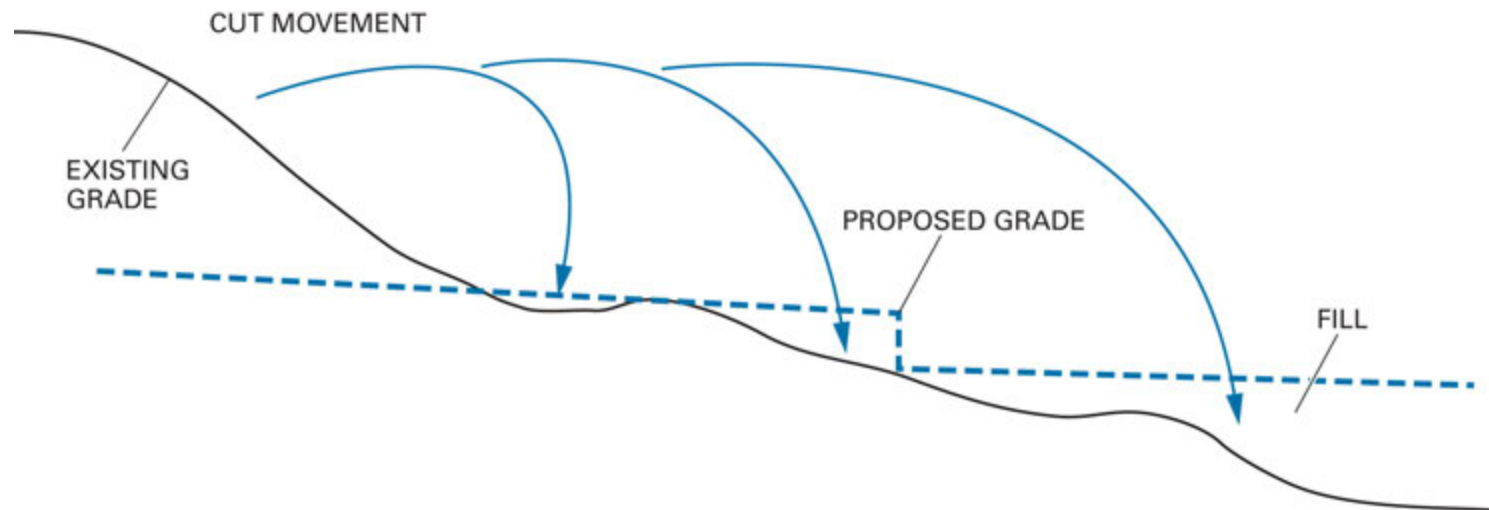
2.1.2 Cycle Times

- Cycle time is the measurement of how long it takes to perform a repetitive operation, such as the time for a piece of equipment to load, haul, dump, and return to the loading point to start the next cycle.
- This allows the contractor to estimate how long it will take to complete a specific activity, like rough grading or excavating a ditch.
- Cycle times vary for different types of equipment.
- Most equipment manufacturers provide information about average cycle times.



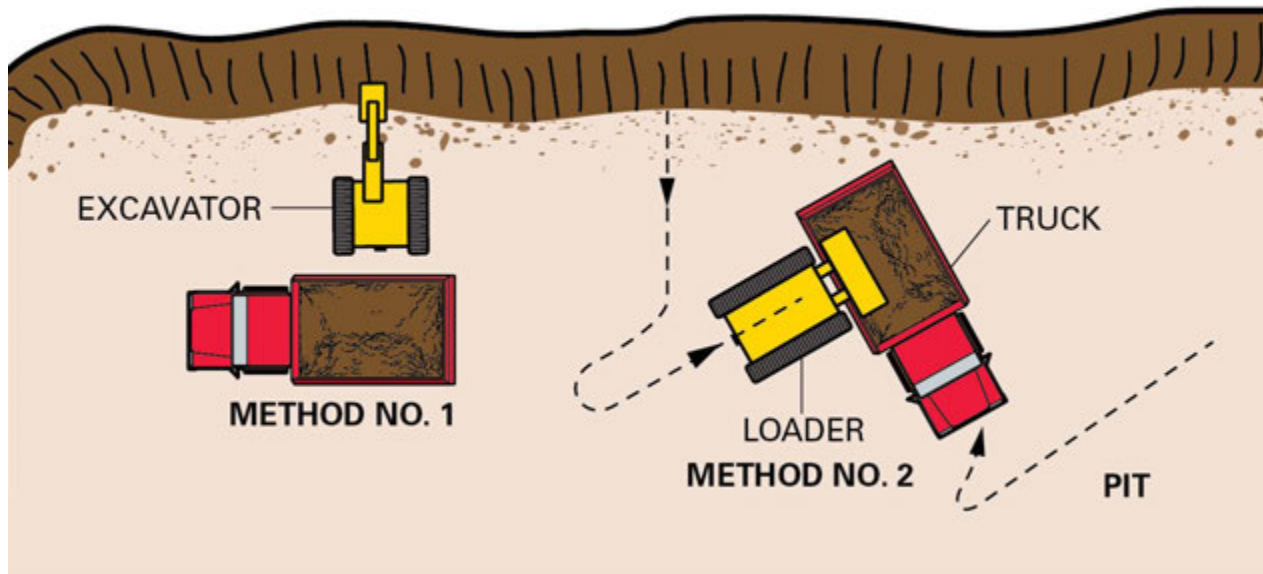
2.1.3 Roadway Excavation

- Roadway excavation is a leveling process used in highway construction and airfield grading.
- Often the soil needed for fill areas is obtained from cut areas and there is no need to haul material in from outside.



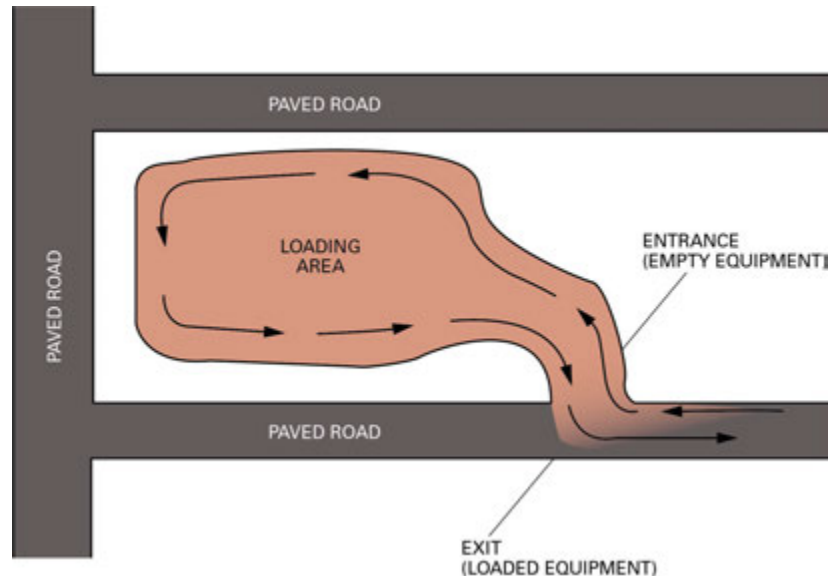
2.1.4 Bulk Pit Excavation (1 of 2)

When excavated material is loaded onto a truck for removal from the site, it is important to minimize the movement of the excavator or loader for the shortest cycle time.



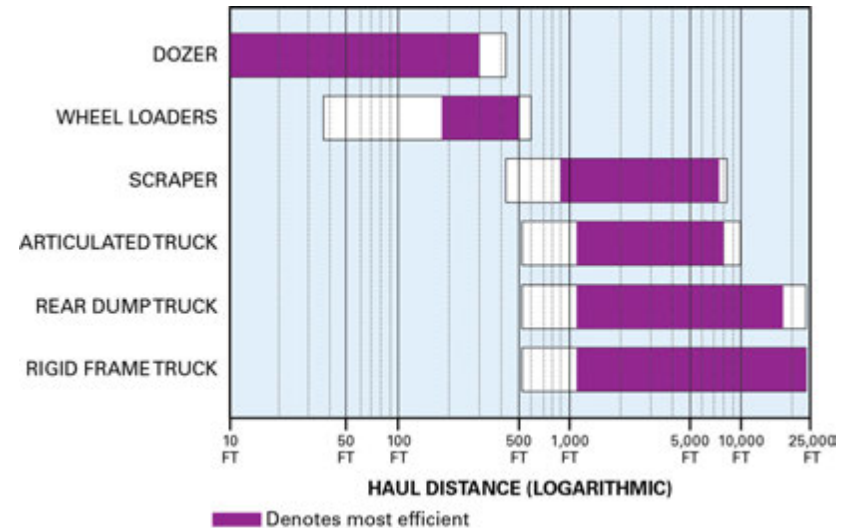
2.1.4 Bulk Pit Excavation (2 of 2)

- In this method, it is only necessary to push the excavated material aside.
- A crawler-mounted loader or dozer uses its bucket or blade to dig and collect material on the front, and then pushes beyond the access ramp to the fill site.



2.1.5 Bulk Wide-Area Excavation

- This type of excavation permits access to the area from many directions.
- Bulk wide-area excavation is used in highway construction, airfield runway grading, removal of top layers of soil, and building of earthen dams and basins.
- Equipment selection depends on the size of the job and the characteristics of the soils being moved.



2.1.6 Channel Excavation

- Channel excavation is the excavation of loose, unconsolidated materials, of material lying under water (such as a drainage channel), or of saturated soils that prevent the equipment from traveling over the excavated area.
- This type of excavation is usually performed by equipment standing on solid ground at the same level or higher than the material being worked.



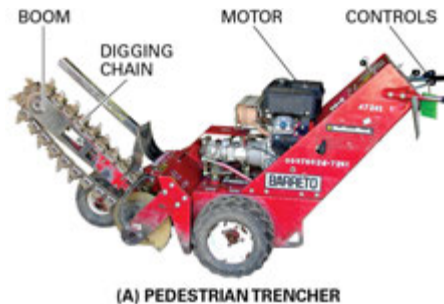
2.1.7 Limited Area Vertical Excavation

- Some excavation must be done by lifting the material vertically out of the pit because the sides of the excavation require shoring or shielding.
- This type of excavation is known as limited-area vertical excavation.



2.1.8 Trench Excavation

A trench should not be excavated wider than is necessary. Shoring can be used to reinforce the sides of the trench.



2.2.0 Loading Excavated Soil

- Loading is the first step in the earthmoving cycle.
- The specific job and the type of material encountered has a great influence on both the selection of equipment and the method used.



2.2.1 Selection of Loading Equipment

Selection of loading equipment depends on factors such as cycle time, type of material, equipment efficiency, and cost.

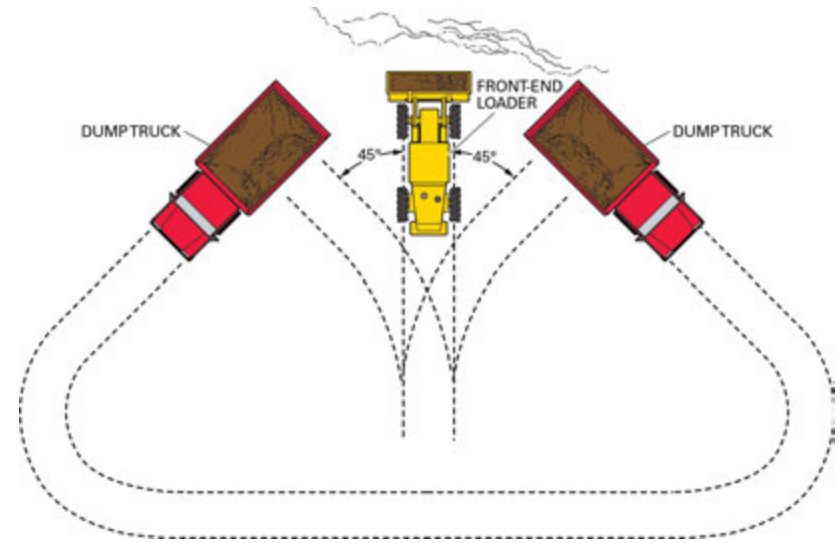
The options available for mobile loading equipment include:

- Front-end loaders (and variations)
- Excavators
- Scrapers
- Shovels
- Other: conveyors, overhead buckets, and cribs



2.2.2 Methods of Loading

- Load from piled material or load from an excavation.
- When loading from stockpiles and spoils, the front-end loader is the primary piece of equipment used.
- Loaders usually follow an I- or Y-pattern when loading from a bank of material.
- If the loader has to travel more than one and a half wheel revolutions, the operation is inefficient.



2.3.0 Hauling Excavated Soil

Hauling is normally broken into two types:

- **Over-the-road hauling** is hauling over public highways using a haul unit such as a dump truck.
- **Off-road hauling** is done within a job site and not on public highways. Equipment used for off-road hauling includes large trucks that are not designed to travel on public highways, as well as scrapers, front-end loaders, and regular trucks.



2.3.1 Selection of Hauling Equipment

- **End-dump (rear dump)** — These types of trucks should be used if material consists of rock fragments or masses of slate or shale that are too large for other units to unload, or if dumping is to be done over the edge of a waste bank or fill.
- **Bottom- or side-dump** — These units are good for transporting free-flowing materials over reasonably level haul routes that permit a high travel speed.
- **Scrapers** — Scrapers can haul a large amount of material at relatively high speeds for short distances. They have an added advantage of being able to load the material directly by cutting or scraping a thin layer of soil while moving forward at a slow speed.



2.3.2 Hauling Methods

- Hauling is the largest single factor in cycle times for earthmoving operations.
- Synchronizing the hauling operation includes the timing of the hauling as well as the traffic pattern.
- A traffic control supervisor may be required for operations using a substantial number of haul units.



2.4.0 Dumping Excavated Soil

- Material may be dumped either to build up an area to make an embankment or to dispose of spoils.
- The dumping must be closely controlled when building up an embankment or when dumping at a spoils site.



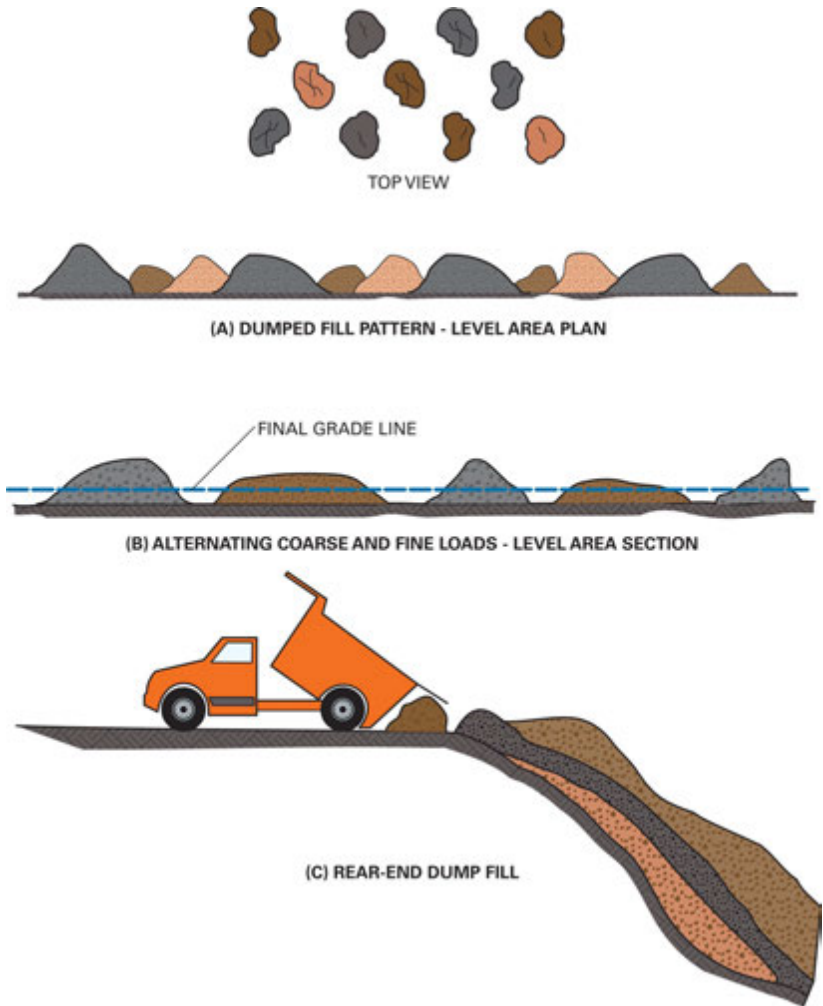
2.4.1 Dump Sites

- Spoils areas are used for the disposal of surplus excavation materials, materials that are unsuitable for fills, or for storage of materials that will be reused.
- The location of the spoils depends on the type of material, the haul distance, and the overall costs.
- If the material to be dumped is for filling, backfilling, or making an embankment, these areas are shown on the plans and indicated with the survey and slope stakes.
- Stockpiles are used to store material for later use in the project.



2.4.2 Dump Patterns

Dump patterns depend on the material being dumped and the type of haul equipment being used.



2.4.3 Return Run

When **backhauling**, equipment operators must:

- Drive cautiously at all times.
- Use the prescribed return route from the dump site.
- Follow markers laid down by the project engineer if driving through a construction site.
- Stay in line and following the directions of the traffic supervisor.
- Think ahead about how to maneuver the machine when approaching the loading site for the next cycle.



2.5.0 Fill, Backfill, and Embankment Construction

- Excavated materials can often be used as fill, backfill, or to build an embankment.
- Materials must be tested according to local regulations before being used for these purposes.



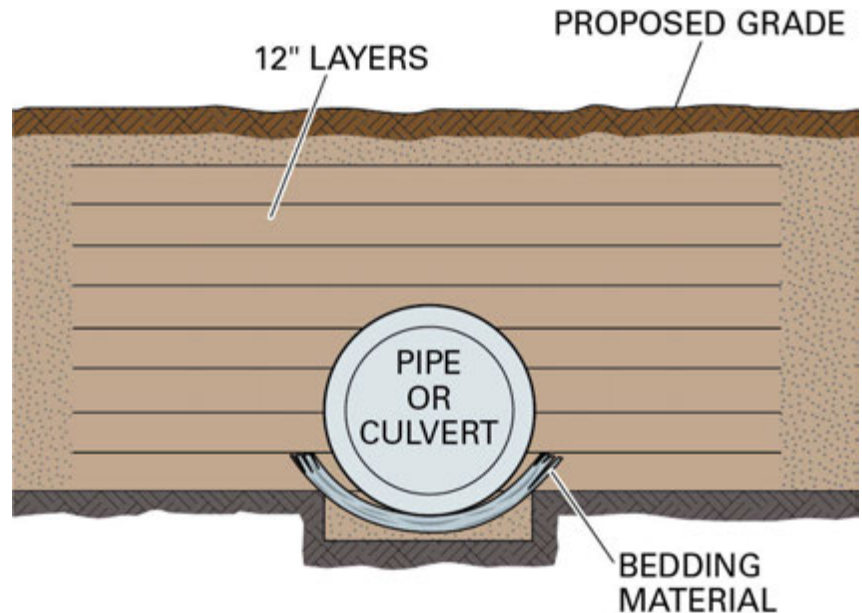
2.5.1 Fill Material

- **Common borrow fill** — Common borrow fill is brought onto the construction site and dumped in piles. It is then spread uniformly over an area and compacted according to the job specifications. It can be used to build up an area where little or no compaction is required.
- **Waste fill** — Waste fill is recycled, treated material approved for use in certain landfill areas. Compaction requirements are based on where the material is being placed.
- **Select fill** — Select fill is material with specific properties that can be obtained only by selective excavation or manufacture. It is usually more expensive than other types of fill and must be used, placed, and compacted as specified by the project plans.



2.5.2 Backfill

- Backfill is the returning of excavated material to an excavation site.
- It is important to backfill and use the appropriate compaction method so that a minimal amount of future settling occurs.
- For backfilling around small structures and buried utilities, a backhoe is the ideal piece of equipment.

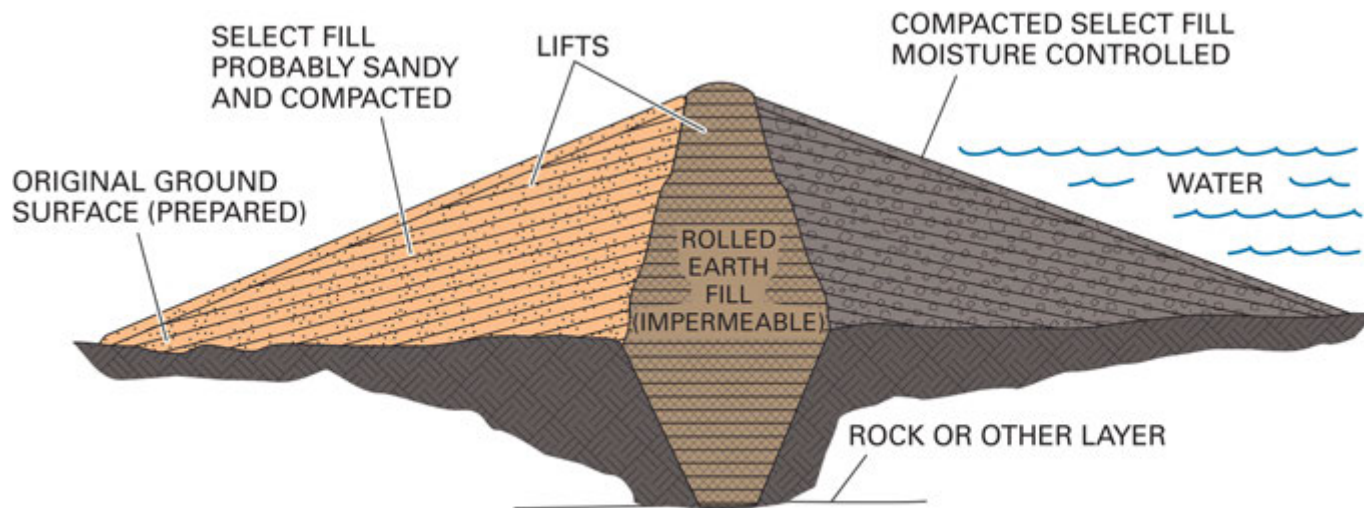


- Use fill that will compact adequately.
- Sift fill into trench in 12" layers to prevent air pockets.
- Bury pipe or culvert at least 2'.
- Compact and fill as necessary to achieve proposed grade.

2.5.3 Embankment Construction (1 of 2)

Embankments are fills that are placed for specific purposes under carefully controlled conditions.

They are classified as either equipment-compacted fill or rolled-earth fill.



- Fill on both sides of rolled earth will provide it with structural support for added strength.

2.5.3 Embankment Construction (2 of 2)

- Embankment material is spread and leveled in layers or **lifts**.
- The equipment and procedures involved depend on equipment availability and the type of material being used.
- Fill should be placed evenly.
- Each lift should be spread to a uniform thickness that does not exceed the specified thickness before compaction.
- Each lift must be compacted before the next lift is placed.
- Where large chunks of material are placed, extra compaction and blading are needed to spread the material uniformly.



Wrap Up – Trade Terms (1 of 2)

Backhauling: The return trip of a piece of equipment after it has completed dumping its load.

Consolidation: To become firm by compacting soil particles so they are closer together.

Gradation: The classification of soils into different particle sizes.

Impermeable: The property of a material that does not allow movement of a fluid through it; for example, an impermeable soil will not allow liquid water to pass through it.

Lifts: When referring to soil compaction during construction, layers of soil that are placed in a fill area or on an embankment before compaction.



Wrap Up – Trade Terms (2 of 2)

Pay item: A defined piece of material or work that the contractor is paid for. Pay items are usually expressed as unit costs.

Spoils: Excavated materials reserved for future use or unsuitable for construction that have been stockpiled.

Windrows: Long, straight piles of placed material for the purpose of mixing or scraping up.

2.0.0 Section Review Question 1

1. Bulk wide-area excavations occur mainly for the following *except* _____.
- a. highway construction
 - b. strip mining
 - c. earthen dam projects
 - d. building foundation preparation



2.0.0 Section Review Question 1 Answer

1. Bulk wide-area excavations occur mainly for the following *except* ____.
- a. highway construction
 - b. strip mining
 - c. earthen dam projects
 - d. building foundation preparation**



2.0.0 Section Review Question 2

2. The mobile equipment suitable for loading soil at excavations includes _____.
- a. mobile cranes
 - b. front-end loaders
 - c. stiff-leg derricks
 - d. rigid-frame dump trucks



2.0.0 Section Review Question 2 Answer

2. The mobile equipment suitable for loading soil at excavations includes ____.
- a. mobile cranes
 - b. front-end loaders**
 - c. stiff-leg derricks
 - d. rigid-frame dump trucks



2.0.0 Section Review Question 3

3. The type of equipment best suited for hauling large rock fragments or masses of slate/shale is the _____.

- a. end-dump truck
- b. scraper
- c. bottom-dump truck
- d. scarifier



2.0.0 Section Review Question 3 Answer

3. The type of equipment best suited for hauling large rock fragments or masses of slate/shale is the _____.

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- b. scraper
- c. bottom-dump truck
- d. scarifier



2.0.0 Section Review Question 4

4. For ease of blending different types of materials that do not blend easily, such as clay and sand, the best dumping pattern is _____.
- a. dumping the materials in alternating piles on a level area
 - b. dumping the materials in alternating windrows
 - c. dumping and then spreading a layer of one material, and then the same for the other
 - d. dumping the materials in alternating layers



2.0.0 Section Review Question 4 Answer

4. For ease of blending different types of materials that do not blend easily, such as clay and sand, the best dumping pattern is _____.
- a. dumping the materials in alternating piles on a level area
 - b. dumping the materials in alternating windrows
 - c. dumping and then spreading a layer of one material, and then the same for the other**
 - d. dumping the materials in alternating layers



2.0.0 Section Review Question 5

5. What kind of fill tends to separate by particle sizes when dumped down a steep slope into a deep depression?
- a. Select fill
 - b. Common borrow fill
 - c. Waste fill
 - d. Consolidated fill



2.0.0 Section Review Question 5 Answer

5. What kind of fill tends to separate by particle sizes when dumped down a steep slope into a deep depression?
- a. Select fill
 - b. Common borrow fill**
 - c. Waste fill
 - d. Consolidated fill



Next...

Section 3.0.0 Stabilizing Soils



*Read Sections 3.0.0 through 3.3.0 and
complete the Section Review.*